Project Euler Problem 6: Sum Square Difference

Brinhasavlin

9/30/2026

Problem

The sum of the squares of the first ten natural numbers is,

$$1^2 + 2^2 + \cdots + 10^2 = 385.$$

The square of the sum of the first ten natural numbers is,

$$(1 + 2 + \cdots + 10)^2 = 55^2 = 3025.$$

Hence, the difference between the sum of the squares of the first ten natural numbers and the square of the sum is $3025 - 385 = 2640$.

Find the difference between the sum of the squares of the first one hundred natural numbers and the square of the sum.

Proof

This solution will derive from two known properties. There is a closed form for both the sum of squared integers and the squared sum of integers.

Lemma 1. *For all $n \in \mathbb{N}$,*

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

Proof. We proceed by induction on n.

Base case ($n = 1$). The left-hand side is $\sum_{k=1}^{1} k = 1$, and the right-hand side is

$$\frac{1(1+1)}{2} = \frac{2}{2} = 1,$$

so the statement holds for $n = 1$.

Inductive step. Suppose that for some $n \geq 1$,

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}. \tag{IH}$$

We show the statement holds for $n+1$:

$$\begin{aligned}
\sum_{k=1}^{n+1} k &= \sum_{k=1}^{n} k + (n+1) && \text{(split off last term)} \\
&= \frac{n(n+1)}{2} + (n+1) && \text{by (IH)} \\
&= \frac{n(n+1) + 2(n+1)}{2} \\
&= \frac{(n+1)(n+2)}{2}.
\end{aligned}$$

Hence, the statement holds for all $n \in \mathbb{N}$. □

Lemma 2. *For all $n \in \mathbb{N}$,*

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Proof. We proceed by induction on n.

Base case ($n = 1$). The left-hand side is $\sum_{k=1}^{1} k^2 = 1^2 = 1$, and the right-hand side is

$$\frac{1(1+1)(2 \cdot 1 + 1)}{6} = \frac{6}{6} = 1,$$

so the statement holds for $n = 1$.

Inductive step. Suppose that for some $n \geq 1$,

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}. \tag{IH}$$

We show the statement holds for $n+1$:

$$\begin{aligned}
\sum_{k=1}^{n+1} k^2 &= \sum_{k=1}^{n} k^2 + (n+1)^2 && \text{(split off last term)} \\
&= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 && \text{by (IH)} \\
&= \frac{n(n+1)(2n+1) + 6(n+1)^2}{6} \\
&= \frac{(n+1)\big(n(2n+1) + 6(n+1)\big)}{6} \\
&= \frac{(n+1)(2n^2 + 7n + 6)}{6} \\
&= \frac{(n+1)(n+2)(2n+3)}{6}.
\end{aligned}$$

Hence, the statement holds for all $n \in \mathbb{N}$. □

Theorem 3. *For all $n \in \mathbb{N}$,*

$$\left(\sum_{k=1}^{n} k\right)^2 - \sum_{k=1}^{n} k^2 = \frac{n(n^2-1)(3n+2)}{12}.$$

Proof. By lemmas 1 and 2,

$$\begin{aligned}
\left(\sum_{k=1}^{n} k\right)^2 - \sum_{k=1}^{n} k^2 &= \left(\frac{n(n+1)}{2}\right)^2 - \frac{n(n+1)(2n+1)}{6} \\
&= \frac{3\big(n(n+1)\big)^2 - 2n(n+1)(2n+1)}{12} \\
&= \frac{n(n+1)\big(3n(n+1) - 2(2n+1)\big)}{12} \\
&= \frac{n(n+1)(3n^2-n-2)}{12} \\
&= \frac{n(n+1)(n-1)(3n+2)}{12} \\
&= \frac{n(n^2-1)(3n+2)}{12}.
\end{aligned}$$

□

Answer

Applying the theorem with $n = 100$,

$$\begin{aligned}
\left(\sum_{k=1}^{100} k\right)^2 - \sum_{k=1}^{100} k^2 &= \frac{100(100^2-1)(3\cdot 100+2)}{12} \\
&= \frac{100 \cdot 9999 \cdot 302}{12} \\
&= \frac{301969800}{12} \\
&= 25164150.
\end{aligned}$$